OCC: SOFTWARE ENGINEER YEAR 2

SAQA ID
NQF 6
Software design and testing C#
SDT621
WEEK 2 FRIDAY PROJECT
2026

CTU training solutions | Private Tertiary Education

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DECLARATION OF AUTHENTICITY

I _____ (GROUP MEMBER NUMNER)

hereby declare that the contents of this assessment _____

is entirely my own work, completed by me without any paraphrasing/copying, or presented as my own work accessed from any AI Apps, example ChatGPT, Co-pilot, Perplexity, or any other App.

I have read and understood the Examination Rules and Regulations.

Signature: _____

Date: _____

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QUALIFICATION AND PROJECT DETAILS

Programme Title	OCC: SOFTWARE ENGINEER
Module Title	Software design and testing C#
Module Code	SDT621
NQF Level	6
Assessment Type	WEEK 2 Friday PROJECT-2
Hand Out Date & Time	@09h30am - 24 / 0 4 / 2026
Hand In & Time	@11h00AM - 24 April 2026
Project Lead Developer	Lusukama Selemani
Submission code via	Github and share the link

Student Name and Surname	Group Member Number

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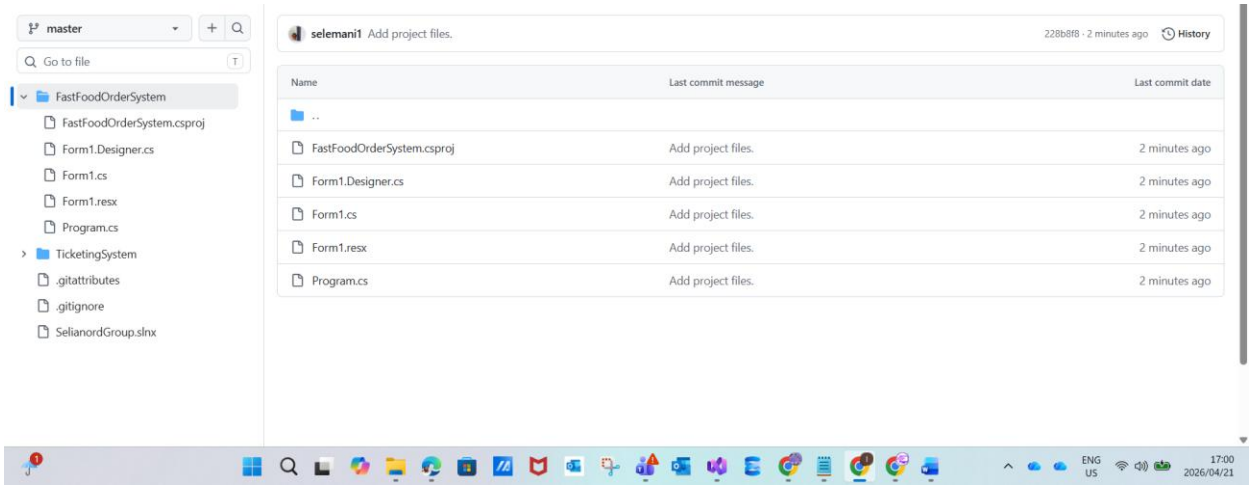
Example of Collecting Evidence for your submission pdf:

Question1 :



Ensure each screenshot shows the **full screen**, including the **system date and time**

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Group Project: Ticket Application System (Windows Forms + GitHub Collaboration)

Project Type

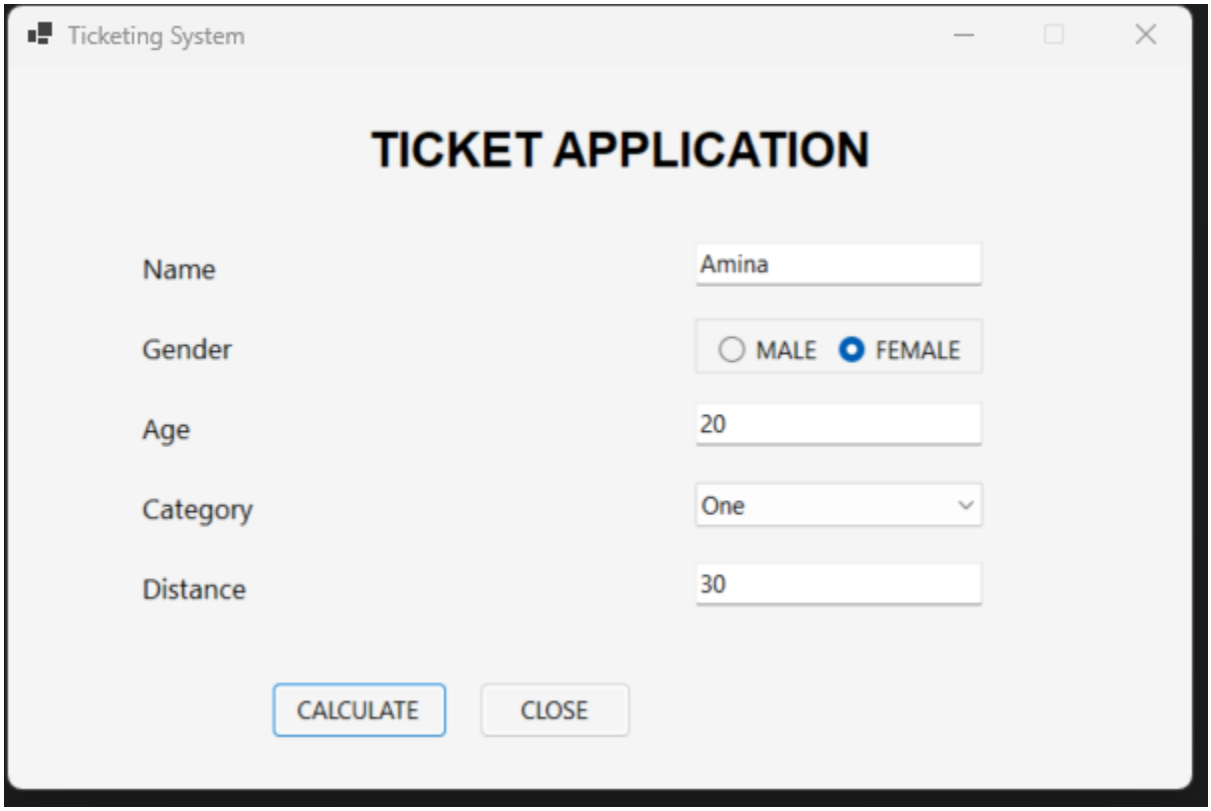
Group Software Development Project

Minimum Group Members

4 students per group

Each member **must have a clearly defined role** and contribute using **GitHub version control**.

Project Overview



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Your group must design and develop a **C# Windows Forms Application** called:

Ticket Application System

The system allows a user to:

- enter passenger name
- select gender
- enter age
- select travel category
- enter travel distance
- calculate ticket price
- apply discount rules
- display final ticket summary

The system must be developed collaboratively using **GitHub**.

Functional Requirements

The application must contain the following controls:

Input Fields

TextBox

- Passenger Name
- Age
- Distance

RadioButtons

Gender selection:

- Male
- Female

ComboBox

Category selection:

- Category One
- Category Two
- Category Three

Buttons

- Calculate Ticket Price
- Clear Form
- Exit Application

Output

Display ticket summary using:

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MessageBox OR Summary Panel

Business Rules

The system must apply the following pricing rules:

Category Pricing

Category	Price per km
One	R20
Two	R35
Three	R50

Discount Rule

If passenger age is:

Less than 12 years
→ Ticket is FREE

Gender Rule

If passenger is **Female**
→ Apply **50% discount**

Non-Functional Requirements

The application must:

- have a professional user interface
- validate user input
- prevent invalid entries
- follow correct naming conventions
- include comments in code
- be tested before submission
- use GitHub collaboration properly

Group Member Roles (Minimum 4 Members)

Each member must contribute separately using GitHub.

Member 1: Project Manager / GitHub Coordinator

Responsibilities:

- create GitHub repository
- invite group members
- create project structure
- manage branches
- merge final code

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- maintain README file
- coordinate submission

Deliverables:

- repository setup
- commit structure
- branch management
- final integration

Member 2: UI Developer

Responsibilities:

Design the Windows Form interface

Add:

- labels
- textboxes
- radiobuttons
- combobox
- buttons
- layout alignment

Improve:

- spacing
- fonts
- usability
- professional appearance

Deliverables:

- working interface
- properly named controls
- responsive layout

Member 3: Logic Developer

Responsibilities:

Write calculation logic for:

- Category pricing
- Category One → 20 × distance
- Category Two → 35 × distance
- Category Three → 50 × distance

Apply rules:

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Age < 12 → free ticket

Female → 50% discount

Implement:

- calculate button logic
- clear button logic
- exit button logic

Deliverables:

- working calculation system
- discount logic
- validation logic

Member 4: Tester and Documentation Specialist

Responsibilities:

Test the application

Verify:

- calculations correct
- discount rules applied correctly
- inputs validated
- buttons working correctly

Prepare documentation:

- screenshots
- test cases
- results table
- GitHub contribution evidence

Deliverables:

- test plan
- bug report
- screenshots
- final documentation support

GitHub Collaboration Requirements

Each group **must use GitHub**

GitHub Workflow

One student creates repository:

Example:

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TicketApplicationSystem

Add collaborators

Each member creates their own branch:

Example:

- Dev-ui-design-YourName
- Dev-logic-module-YourName
- Dev-testing-docs-YourName
- Dev-project-management-YourName

Example Commit Messages

Good examples:

- Created Windows Form layout
- Added gender selection controls
- Implemented category pricing logic
- Applied female discount rule
- Added validation for distance input
- Created test cases documentation

Avoid:

- update
- changes
- fixed stuff

Required GitHub Evidence

Repository must include:

- README file
- Source code
- Commit history from ALL members
- Branches used
- Screenshots folder
- Documentation file

Project Tasks Summary

Task 1: Design User Interface

Create Windows Form with:

- Name textbox
- Gender radiobuttons
- Age textbox
- Category combobox

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- Distance textbox
- Calculate button
- Clear button
- Exit button

Task 2: Implement Pricing Logic

System must calculate:

- price = category_rate × distance
- Then apply:
- if age < 12 → price = 0
- if female → price = price × 0.5

Task 3: Validation Rules

System must check:

- Name not empty
- Age numeric
- Distance numeric
- Category selected
- Gender selected

Task 4: Display Ticket Summary

Display:

- Passenger name
- Gender
- Age
- Category
- Distance
- Final ticket price

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Deliverables

Each group must submit:

1 GitHub Repository Link

Working repository link

Must include commits from all members

3 Project Report

Include:

- Project title
- Group members
- Student numbers
- Member roles
- Screenshots
- Test results
- GitHub activity evidence
- Challenges faced
- Conclusion

4 Contribution Declaration

Each student must clearly state:

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Student Name	What they implemented	Which files they modified	Which commits they made	Status